

1. 解: $A = -2$ $B = 1$ $Q = 1$ $R = 1$

由里卡蒂微分方程得:

最优控制力 $u^*(t) = -R^{-1}B^T P x = -P x$

代数里卡蒂微分方程:

$$-PA - A^T P + P B R^{-1} B^T P - Q = 0$$

$$\text{即 } 02P + 2P^2 + P^2 - 1 = 0$$

$$P_1 = -2 + \sqrt{5} \quad P_2 = -2 - \sqrt{5} \text{ (舍)}$$

$$\text{则 } P = -2 + \sqrt{5} \quad u^*(t) = (2 - \sqrt{5}) x$$

2. 解: $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $Q = \begin{bmatrix} 1 & 0 \\ 0 & m \end{bmatrix}$ $R = n$

最优控制力 $u^*(t) = -R^{-1}B^T P x = -\frac{1}{n} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

设 $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$ $\begin{pmatrix} P_{12} = P_{21} \end{pmatrix}$ $= -\frac{1}{n} (P_{21} x_1 + P_{22} x_2)$

代数里卡蒂微分方程:

$$-PA - A^T P + P B R^{-1} B^T P - Q = 0$$

$$\text{即 } -\begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} +$$

$$\begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{1}{n} \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & m \end{bmatrix} = 0$$

$$-\begin{bmatrix} 0 & P_{11} \\ 0 & P_{21} \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ P_{11} & P_{12} \end{bmatrix} + \frac{1}{n} \begin{bmatrix} P_{12} P_{21} & P_{12} P_{22} \\ P_{22} P_{21} & P_{22} P_{22} \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & m \end{bmatrix} = 0$$

又 $P_{12} = P_{21}$, 可得:

$$-\frac{1}{n} P_{12}^2 - 1 = 0$$

$$-P_{11} + \frac{1}{n} P_{12} P_{22} = 0$$

$$-P_{11} + \frac{1}{n} P_{12} P_{22}$$

$$-2P_{12} + \frac{1}{n} P_{22}^2 - m = 0$$

$$P_{12} = \sqrt{n}$$

$$P_{11} = \frac{\sqrt{2n\sqrt{n}+mn}}{n} \sqrt{2\sqrt{n}+m}$$

$$P_{22} = \sqrt{2n\sqrt{n}+mn}$$

$$\text{则 } u^*(t) = -\frac{1}{n} (\sqrt{n} x_1 + \sqrt{2n\sqrt{n}+mn} x_2) = -\frac{\sqrt{n}}{n} x_1 - \frac{\sqrt{2n\sqrt{n}+mn}}{n} x_2$$

(2) 由(1)中 $u^*(t)$ 的结果可得到闭环系统

$$\dot{x} = (A - B R^{-1} B^T P) x$$

即 $\dot{x} = \begin{bmatrix} 0 & 1 \\ -\frac{\sqrt{n}}{n} & -\frac{\sqrt{2n\sqrt{n}+mn}}{n} \end{bmatrix} x$ 令 $A^* = \begin{bmatrix} 0 & 1 \\ -\frac{\sqrt{n}}{n} & -\frac{\sqrt{2n\sqrt{n}+mn}}{n} \end{bmatrix}$

特征方程:

$$|\lambda I - A^*| = \begin{vmatrix} \lambda & -1 \\ \frac{\sqrt{n}}{n} & \lambda + \frac{\sqrt{2n\sqrt{n}+mn}}{n} \end{vmatrix} = \lambda^2 + \frac{\sqrt{2n\sqrt{n}+mn}}{n} \lambda + \frac{\sqrt{n}}{n} = 0$$

$$\lambda_1 = \frac{-\sqrt{mn+2n\sqrt{n}} - \sqrt{mn-2n\sqrt{n}}}{2n}$$

$$\lambda_2 = \frac{-\sqrt{mn+2n\sqrt{n}} + \sqrt{mn-2n\sqrt{n}}}{2n}$$

最优闭环极点为 λ_1, λ_2

(3) 若闭环极点为 $-2, -2$

则 $\lambda_1 = \lambda_2 = -2$

即 $\begin{cases} \frac{-\sqrt{mn+2n\sqrt{n}} - \sqrt{mn-2n\sqrt{n}}}{2n} = \frac{-\sqrt{mn+2n\sqrt{n}} + \sqrt{mn-2n\sqrt{n}}}{2n} \\ \frac{-\sqrt{mn+2n\sqrt{n}} - \sqrt{mn-2n\sqrt{n}}}{2n} = -2 \end{cases}$

化简得: $\begin{cases} \sqrt{mn-2n\sqrt{n}} = 0 \\ \sqrt{mn+2n\sqrt{n}} = 4n \end{cases} \quad \begin{cases} m = \frac{1}{2} \\ n = \frac{1}{16} \end{cases}$

(4) 若 $m=0$, 闭环极点为 $\sqrt{2}(-1 \pm i)$

$$\lambda_1 = \frac{-\sqrt{2n\sqrt{n}} - \sqrt{-2n\sqrt{n}}}{2n} \quad \lambda_2 = \frac{-\sqrt{2n\sqrt{n}} + \sqrt{-2n\sqrt{n}}}{2n}$$

~~若~~ $-\sqrt{2} = \frac{-\sqrt{2n\sqrt{n}}}{2n}$ 得: $n = \frac{1}{16}$

验证: $\frac{\sqrt{2n\sqrt{n}}}{2n} = \sqrt{2}i$, λ_1, λ_2 为 $\sqrt{2}(-1 \pm i)$, 正确

则 $n = \frac{1}{16}$